

ECE 759 High Performance Computing for Engineering Applications Assignment 1

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My GitHub link: <https://github.com/BillWan-zzzyyy/repo759/tree/main/HW01>

1.

- (a) Read the files `timing.md` and `slurm_usage.md` from the `Assignments/general` directory of the ECE 759 Resource Repo. These documents will come into play in almost all your assignments throughout the semester.
- (b) Read the `hw_repos.md` file in the same directory. This is very important and must be done in order for you to turn in all the assignments for ECE 759.
- (c) At least skim `workflow.md`. This contains a quick guide for effectively working between your local computer and Euler. This was also discussed in class.

Answer: I went through a) through c) and understand how to time code, how to submit my assignments with `git`, and what the recommended workflow is when it comes to working on my assignment

2.

Write one line of `bash` code for each of the following sub-tasks (assume that all the files and directories mentioned exist).

- (a) Change the current directory to a subdirectory called `somedir`.

Answer: `cd somedir`

- (b) Print out to the terminal the contents of a file called `sometext.txt`. The file exists in the current directory.

Answer: `cat sometext.txt`

- (c) Print out to the terminal the last 5 lines of a plain text file called `sometext.txt`. The file exists in the current directory.

Answer: `tail -n 5 sometext.txt`

- (d) Print out to the terminal the last 5 lines of *each* file that ends in the extension `.txt` and lives in the current directory.

Answer: `tail -n 5 -- *.txt`

- (e) Write a `for` loop which prints each integer from 0 to 6 (including 0 and 6).

Answer: `for i in {0..6}; do echo "$i"; done`

3.

The purpose of this task is to get you familiar using *Euler*. On Euler, using the `module` command, answer the following questions.

- (a) Are there any modules loaded (`module list`) when you log in on Euler?

Answer: No modules loaded

- (b) What version (version number) of `gcc` is available to you without loading any modules?

Answer: `gcc (GCC) 14.3.1 20251022 (Red Hat 14.3.1-4) Copyright (C) 2024 Free Software Foundation, Inc.`

- (c) List all `cuda` modules available on Euler.

Answer: `nvidia/cuda/10.2.2`
`nvidia/cuda/11.0.3`
`nvidia/cuda/11.3.1`
`nvidia/cuda/11.6.0`
`nvidia/cuda/11.8.0`
`nvidia/cuda/12.0.0`
`nvidia/cuda/12.1.0`
`nvidia/cuda/12.2.0`
`nvidia/cuda/12.5.0`
`nvidia/cuda/12.9.1`
`nvidia/cuda/13.0.0 (D)`
`nvidia/nvhpc-hpcx-cuda11/24.5`
`nvidia/nvhpc-hpcx-cuda12/23.11`
`nvidia/nvhpc-hpcx-cuda12/24.5 (D)`

- (d) List one other piece of software that has a module on Euler and write one sentence about what it does. (If you aren't familiar with any of the other software, google one up and write a sentence about it.)

Answer: abaqus/2023. Abaqus is a commercial finite element analysis software package used to simulate structural, thermal, contact, and other engineering problems.

4.

Write a bash script called `task4.sh` with a Slurm header which asks for

- 2 CPU cores
- A job name of `FirstSlurm`
- An output file called `FirstSlurm.out`
- An error file called `FirstSlurm.err`

and runs a single command to print the hostname of the machine (compute node) running the job. This job must be submitted by running `sbatch task4.sh` on Euler's head node.

Answer: Submit the following source separately as `HW01/task4.sh` in the GitHub repository:

```
#!/bin/bash

#SBATCH -p instruction
#SBATCH -c 2
#SBATCH -J FirstSlurm
#SBATCH -o FirstSlurm.out
#SBATCH -e FirstSlurm.err

hostname
```

5.

Research some useful Slurm tools (one sentence responses):

- (a) In what directory does a Slurm job on Euler begin execution? You may run some jobs in different directories to check this.

Answer: A Slurm job begins in the directory from which it was submitted, unless the script or submission specifies another directory.

- (b) Explain what `SLURM_JOB_ID` is in the environment of a running Slurm job.

Answer: `SLURM_JOB_ID` is the unique identifier Slurm assigns to the running job.

- (c) How would you track the status of job(s) run by yourself? Assume that the job(s) have not been completed yet.

Answer: Run `squeue -u "$USER"` to list the current status of queued or running jobs.

- (d) How would you cancel a job submitted by yourself? Assume that the job is still in the queue.

Answer: Run `scancel JOB_ID` with the job's identifier to cancel it.

- (e) Explain what the following script header line specifies: `#SBATCH --gres=gpu:1`

Answer: It requests one GPU as a generic resource for each allocated node of the job.

- (f) (Optional) Explain what the following script header line specifies:

`#SBATCH --array=0-9`

Answer: It creates ten array tasks with indices 0 through 9.

6.

Write a C++ program called `task6.cpp` that:

- (a) Takes a command line argument `N`. (If you are confused about command line arguments, it may be helpful for you to read this)
- (b) Prints out each integer from 0 to `N` (including 0 and `N`) in ascending order with the `printf` function.
- (c) Prints out each integer from `N` to 0 (including `N` and 0) in descending order with `std::cout`.

Answer:

```
#include <cerrno>
#include <climits>
#include <cstdio>
#include <cstdlib>
#include <iostream>

int main(int argc, char* argv[]) {
    if (argc != 2) {
        std::cerr << "Usage: " << argv[0] << " N\n";
        return 1;
    }

    char* end = nullptr;
    errno = 0;
```

```

const long parsed = std::strtol(argv[1], &end, 10);
if (errno != 0 || end == argv[1] || *end != '\0' || parsed < 0 ||
    parsed > INT_MAX) {
    std::cerr << "N must be a non-negative integer.\n";
    return 1;
}

const int n = static_cast<int>(parsed);

for (int i = 0;; ++i) {
    std::printf("%d%c", i, i == n ? '\n' : ' ');
    if (i == n) {
        break;
    }
}

for (int i = n;; --i) {
    std::cout << i << (i == 0 ? '\n' : ' ');
    if (i == 0) {
        break;
    }
}

return 0;
}

```